

Multivariate

$$Y = AX + b = g(X) \Rightarrow X = A^{-1}(Y - b) = h(Y)$$

$$\text{cov}[Y] = A \text{cov}[X] \cdot A^T$$

$$E[Y] = A \cdot E[X] + b$$

$$J = A^{-1}$$

$$f_X(\bar{x}) = \frac{1}{(2\pi)^{n/2} \cdot |\Sigma|} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

$$\text{cov}[X, Y] = E[(X - E[X])(Y - E[Y])^T]$$

$$\text{Var}[aX + bY] = a^2 \text{Var}[X] + b^2 \text{Var}[Y] + 2ab \text{cov}[X, Y]$$

$$\mu_{a|b} = \mu_a + \Sigma_{ab} \Sigma_{ba}^{-1} (\mu_b - \mu_b)$$

$$\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab} \Sigma_{bb}^{-1} \Sigma_{ba}$$

$$\begin{bmatrix} x_a \\ x_b \end{bmatrix} \sim N \left(\begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix}, \begin{bmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{bmatrix} \right)$$

$$J = \begin{bmatrix} \frac{\partial H_1}{\partial y_1} & \frac{\partial H_1}{\partial y_2} \\ \frac{\partial H_2}{\partial y_1} & \frac{\partial H_2}{\partial y_2} \end{bmatrix}$$

$$f_Y(y) = f_X(h(y)) \cdot |J|$$

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

$$\frac{x_1 + x_2 + \dots + x_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d}$$

ACCEPT-REJECT :

Generate $x \sim q(\cdot)$

Generate $u \sim U[0, 1]$

If $u \leq \frac{p(x)}{c \cdot q(x)} \Rightarrow \text{ACCEPT}$
 Tight bound

Else, Reject & Generate other sample (x)

MGF :

$$M_X(t) = E[e^{tX}]$$

$$E[Z] = E_N[E[Z|N=n]]$$

$$E[Z] = E_N[E[Z|N=n]]$$

$$M_X'(0) = E[X]$$

$$M_X''(0) = E[X^2]$$

$$M_X^{(r)}(t) = E[e^{tX} X^r]$$

$$D_X = \{t \mid M_X(t) < \infty\}$$

Region of Convergence

Probability Convergence

(1) Borel-Cantelli-Lemma

$\forall \epsilon > 0 :$

$$\sum_{n=1}^{\infty} P(|X_n - x| > \epsilon) < \infty$$

$$\therefore X_n \xrightarrow{a.s.} X$$

$$\sum_{n=1}^{\infty} P(X_n(\omega) \neq 0) < \infty$$

$$X_n \xrightarrow{a.s.} X$$

$$(2) X_n \xrightarrow{P} X : (\forall \epsilon > 0)$$

$$\lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) = 0$$

$$(3) X_n \xrightarrow{d} X$$

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

$$(4) X_n \xrightarrow{m.s.} X$$

$$\lim_{n \rightarrow \infty} E[(X_n - X)^2] = 0$$

$$(5) X_n \xrightarrow{L^r} X$$

$$\lim_{n \rightarrow \infty} E[|X_n - X|^r] = 0$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1 \Rightarrow \text{Converges}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} < 1 \Rightarrow \text{Converges}$$

Monte-Carlo

$$\hat{I}_n^g = \frac{1}{n} \sum_{i=1}^n \frac{f(x_i)}{g(x_i)} \leftarrow$$

$$\text{Importance Sampling} \Rightarrow E[h(X)] = E \left[h(X) \frac{f(X)}{g(X)} \right]$$

$$I = \int_0^a f(x) dx \Rightarrow x = au, dx = a du$$

$$x_i = a u_i$$

$$\hat{I}_N^a = \frac{a}{N} \sum_{i=1}^N f(x_i)$$

$$I = \int_0^a f(x) dx \Rightarrow x = \frac{u}{1-u}, dx = \frac{du}{(1-u)^2}$$

$$\hat{I}_N^a = \frac{1}{N} \sum_{i=1}^N g(u_i)$$

Covariance :

$$\text{Cov}[X, Y] = E[XY] - E[X]E[Y]$$

$$\text{cov}[aX, Y] = a \cdot \text{cov}[X, Y]$$

$$\text{cov}[X+a, Y] = a \text{cov}[X, Y]$$

$$\text{cov}[X, Y] = \text{cov}[Y, X]$$

min :

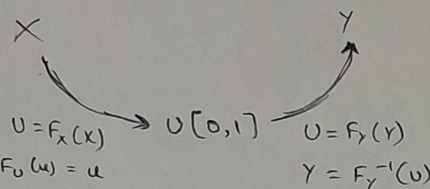
$$P(Y_n \leq y) = 1 - P(Y_n > y)$$

$$= 1 - P(X_1 > y) \cdot P(X_2 > y) \cdot \dots \cdot P(X_n > y)$$

$$= 1 - [1 - P(X_1 \leq y)] \cdot \dots \cdot [1 - P(X_n \leq y)]$$

max :

$$P(Y_n \leq y) = P(X_1 \leq y) \cdot P(X_2 \leq y) \cdot \dots \cdot P(X_n \leq y)$$



Multivariate Gaussian :

$$(a) f_{\bar{x}}(\bar{x}) = \frac{1}{\sqrt{2\pi} |\Sigma|} e^{-\frac{1}{2}(\bar{x} - \mu)^T \Sigma^{-1}(\bar{x} - \mu)}$$

$$(b) X = AZ + \mu \quad \& \quad \Sigma = AA^T$$

Std. Norm. for some A & μ

$$(c) \forall a \in \mathbb{R}^n \Rightarrow a^T X \text{ is univariate Gaussian, } N(a^T \mu, a^T \Sigma a)$$

$$\bar{X} \sim N(\bar{\mu}, \bar{\Sigma})$$

$$(1) P_X(z) = \sum_{i=1}^n P(A_i) \cdot P_{X|A_i}(z)$$

$$(2) E[X] = \sum_{i=1}^n P(A_i) \cdot E[X|A_i]$$

$$(3) f_{XY}(x, y) = f_{X|Y}(x|y) \cdot f_Y(y)$$

$$(4) f_X(x) = \int_y f_{X|Y}(x|y) \cdot f_Y(y) dy = \int_y f_{XY}(x, y) dy$$

$$(5) E[X|Y=y] = \int_x x f_{X|Y}(x|y) dx$$

$$(6) E[X] = \int_y f_Y(y) \cdot E[X|Y=y] dy$$

Independence $\Rightarrow$ Uncorrelated

Uncorrelated $\nRightarrow$ Independence

Memoryless Property: (exp)

$$P(X > a+h | X > a) = \frac{e^{-\lambda(a+h)}}{e^{-\lambda a}} = e^{-\lambda h} = P(X > h)$$

Beta(α, β):

$$f(\theta) = \frac{1}{B(\alpha, \beta)} \cdot \theta^{\alpha-1} \cdot (1-\theta)^{\beta-1}$$

$$E[\theta] = \frac{\alpha}{\alpha + \beta}$$

$$\theta_{MAP} = \frac{\alpha - 1}{\alpha + \beta - 2}$$

$$[\alpha, \beta > 1]$$

$$Var[\theta] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

Uniform $[0, a]$:

$$X_{(n)} = \max X_i$$

$$f_{(n)}(x) = \frac{n}{a^n} \cdot x^{n-1}$$

$$E[X_{(n)}] = \frac{n}{n+1} a < a \text{ (B.J.)}$$

$$Var[X_{(n)}] = \frac{na^2}{(n+2)(n+1)^2}$$

Bayesian:

Prior: (Before data)
 $\pi(\lambda)$

Likelihood: $P(D|\lambda)$ ← ML (argmax)
(Data given Param)

Posterior $\propto$ Prior $\times$ Likelihood ← MAP (argmax)
(Belief After Data)

$$f(x|y) \propto f(y|x) \cdot f(x)$$

$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{f_{Y|X}(y|x) \cdot f_X(x)}{f_Y(y)} = \frac{f_{Y|X}(y|x) \cdot f_X(x)}{\int_{\mathcal{X}} f_{Y|X}(y|x) \cdot f_X(x) dx}$$

$$\theta_{CE} = E[\theta|x=z] \leftarrow \text{Posterior Mean}$$

Bias: $E[\hat{\theta}] = \theta \leftarrow \text{Unbiased}$

$$B[\hat{\theta}] = E[\hat{\theta}] - \theta$$

Asymptotically unbiased: $\lim_{n \rightarrow \infty} E[\hat{\theta}_n] = \theta$

Consistency: $\hat{\theta} \xrightarrow{P} \theta$

$$\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| \geq \epsilon) = 0 \quad \forall \epsilon > 0$$

~~Strongly~~ [consistent]

$$\lim_{n \rightarrow \infty} MSE[\hat{\theta}_n] = 0$$

$$MSE[\hat{\theta}_n] = Var[\hat{\theta}_n] + B^2[\hat{\theta}_n]$$

$$\hookrightarrow E[(\hat{\theta}_n - \theta)^2]$$

~~Cons~~ Strongly consistent: $\hat{\theta}_n \xrightarrow{a.s.} \theta$

$$\sum_{n=1}^{\infty} P(|\hat{\theta}_n - \theta| > \epsilon) < \infty$$

$\hat{\theta}_{ML}$: Asymptotically unbiased & Asymptotically consistent

Poisson (λ):

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$E[X] = \lambda$$

$$Var[X] = \lambda$$

$$\hat{\lambda} = \bar{X}$$

Gaussian (μ, σ^2):

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$$

$$E[\hat{\sigma}_{MLE}^2] = \frac{n-1}{n} \sigma^2 \text{ (B.J.)}$$

$$\hat{\sigma}_{Unb.}^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

LR: (Gaussian)

$$E[X|Y] = \mu_X + \rho \sigma_X \frac{Y - \mu_Y}{\sigma_Y}$$

$$Var[X|Y] = \sigma_X^2 (1 - \rho^2)$$

$$\rho = \frac{cov[X, Y]}{\sigma_X \sigma_Y}$$

Gamma(α, β):

$$f(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \cdot \lambda^{\alpha-1} \cdot e^{-\beta\lambda} \quad (\lambda > 0)$$

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx = (z-1)!$$

$$\int_0^\infty x^a e^{-bx} dx = \frac{\Gamma(a+1)}{b^{a+1}}$$

$$\int_0^1 \theta^m (1-\theta)^r d\theta = \frac{m! r!}{(m+r+1)!}$$

$$E\left[\frac{1}{n} \sum_{i=1}^n \frac{x_i}{i}\right] = \frac{1}{n} \cdot \mu \cdot H_n$$

$$H_n \sim \ln(n) + \gamma$$

$$\int_0^\infty e^{-u^2} du = \frac{\sqrt{\pi}}{2}$$

(Prior) (Likelihood)
Gamma - Poisson:

$$\lambda \sim \text{Gamma}(\alpha, \beta)$$

$$\lambda | D \sim \text{Gamma}(\alpha + S, \beta + S)$$

$$\lambda_{MAP} = \frac{S + \alpha - 1}{\beta + n}$$

(Prior) (Likelihood)
Gaussian - Gaussian:

$$X \sim N(\mu_X, \sigma_X^2)$$

$$W \sim N(0, \sigma_W^2)$$

$$Y = X + W$$

$$X|Y \sim N\left(\frac{\sigma_X^2}{\sigma_X^2 + \sigma_W^2} Y, \frac{\sigma_X^2 \sigma_W^2}{\sigma_X^2 + \sigma_W^2}\right)$$

$$\hat{X}_{MMSE} = \frac{\sigma_X^2}{\sigma_X^2 + \sigma_W^2} Y$$

$$MSE = \frac{\sigma_X^2 \sigma_W^2}{\sigma_X^2 + \sigma_W^2}$$

Markov & Chebyshev

$$P(X \geq a) \leq \frac{E[X]}{a}, \quad X > 0$$

$$P(|X_n - \bar{x}| > a) \leq \frac{Var[X_n]}{a^2}$$

Sigma Algebra:

$$\mathcal{F} = \{\emptyset, \Omega, \mathcal{R}_-, \mathcal{R}_+\}$$

$$(i) \phi \in \mathcal{F} \text{ \& \& } \Omega \in \mathcal{F}$$

$$(ii) B \in \mathcal{F} \Rightarrow B^c \in \mathcal{F}$$

$$(iii) A_1 \in \mathcal{F}, A_2 \in \mathcal{F}$$

$$\Rightarrow A_1 \cup A_2 \in \mathcal{F} \text{ \& \& }$$

$$A_1 \cap A_2 \in \mathcal{F}$$

(Prior) (Likelihood)
Uniform - Bernoulli:

$$\theta \sim U[0, 1] \sim B[1, 1]$$

$$\theta | D \sim B(S+1, n-S+1)$$

$$\theta_{MAP} = S/n$$

$$\theta_{ML} \theta_{CE} = \frac{S+1}{n+2} \quad \left(S = \sum_{i=1}^n x_i\right)$$

Likelihoods:

$$\text{Bernoulli: } L(\theta) = \theta^S (1-\theta)^{n-S}$$

$$\hat{\theta}_{MLE} = \frac{S}{n}$$

$$\text{Poisson: } L(\theta) = e^{-n\theta} \theta^S$$

$$\hat{\theta}_{MLE} = \bar{X}$$

$$\text{Gaussian: } L(\mu, \sigma) = -\frac{1}{2\sigma^2} \sum (x_i - \mu)^2$$

$$\hat{\mu}_{MLE} = \bar{X} \text{ \& \& } \hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$$

Markov Chains

$$\pi P = \pi \Rightarrow P^T \pi^T = \pi^T$$

π belongs to $N(P^T - I)$

$$F_{ii} = 1 \Rightarrow \text{Recurrent state}$$

$$F_{ii} < 1 \Rightarrow \text{Transient state}$$

All states: Reachable $\Rightarrow$ Irreducible

$$P^{(n)} = P^n$$

Property:

$$P(X_n = j | X_1 = x_1, X_2 = x_2, \dots, X_{n-1} = x_{n-1}) = P(X_1 = j | X_{1-1} = x_{n-1})$$

Martingale: $\{X_n, n \geq 0\}$

$$E[X_{n+1} | X_1, X_2, \dots, X_n] = X_n$$

Weiner: $\{X(t): t \geq 0\} \forall t > 0, X(t) \sim N(0, t)$

Random Walk: $\{S_n, n \in \mathbb{Z}^+\}$